

Statistical Review and A/B Testing for New York City TLC Project

Executive summary report

Overview

The goal of this project is to predict taxi fares before each ride, with the broader objective of identifying ways to increase revenue for New York City taxi drivers. This phase of the project focuses on analyzing the relationship between total fare amount and payment type.

Problem

Taxi drivers receive different tip amounts, and this phase of the project aims to explore whether the payment type affects total fare. Specifically, it investigates if customers paying by credit card tend to have higher total fares compared to those paying with cash.

Solution

The Automatidata team conducted an A/B test to examine the effect of credit card payments on total fare amounts. The key business insight is that promoting credit card payments could help taxi drivers increase their revenue.

Details

Steps conducted in the A/B test

1. Collected sample data from an experiment in which customers are randomly selected and divided into two groups:
 - a. Customers who are required to pay with credit card.
 - b. Customers who are required to pay with cash. This enables us to draw causal conclusions about how payment method affects fare amount.
2. Computed descriptive statistics to better understand the average total fare amount for each payment method available to the customer.
3. Conducted a two-sample t-test to determine if there is a statistically significant difference in average total fare between customers who use credit cards and customers who use cash.

A/B test results

The test revealed a statistically significant difference in average total fare between credit card and cash customers. Customers paying by credit card had higher total fares than those paying with cash.

Next Steps

The Automatidata team recommends that the New York City TLC encourage credit card payments and develop strategies to promote them. For instance, TLC could place signs in taxis indicating a preference for credit card payments and implement a protocol for drivers to verbally inform customers that credit cards are preferred.